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Annex D

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Messaging Systems and Services

Student Outline

Data Systems Administrator Course
M09CVQ1
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Terminal Learning Objective

1. Given a command's mission, threat assessment, operational environment, operational plans, virtual environment, and personnel, implement messaging systems and services to establish operating systems in accordance with applicable STIGs and local cybersecurity policies in support of the communications plan. (0671-IMPL-1003)

Enabling Learning Objectives

1. Without the aid of reference, identify in writing the different types of messaging solutions, in accordance with 0671-IMPL-1003

2. Without the aid of reference, identify in writing the characteristics of messaging servers, in accordance with 0671-IMPL-1003

3. Without the aid of reference, identify in writing the unified messaging concepts, in accordance with 0671-IMPL-1003

4. Without the aid of reference, identify in writing the requirements of messaging servers on a domain, in accordance with 0671-IMPL-1003

5. Without the aid of reference, identify in writing the messaging server software requirements, in accordance with 0671-IMPL-1003

6. Without the aid of reference, identify in writing the characteristics of database availability group (DAG), in accordance with 0671-IMPL-1003

7. Without the aid of reference, identify in writing the high-availability solutions for messaging servers, in accordance with 0671-IMPL-1003

8. Without the aid of reference, identify in writing the characteristics of administrative tools, in accordance with 0671-IMPL-1003

9. Without the aid of reference, identify in writing the characteristics of an exchange administration center, in accordance with 0671-IMPL-1003

10. Without the aid of reference, identify in writing the characteristics of messaging clients, in accordance with 0671-IMPL-1003

Different types of Messaging Solutions

What is a Messaging Solution?

Messaging solutions are a type of technology that helps administrators overcome the above challenges through asynchronous, program-to-program communication. Messaging is vital to any network as it provides a way to stay up to date on all relevant information. Messaging is convenient, provides context, and allows conversations to be more personal with emoji's, gifs, and links to information.

Messaging enables software applications to connect and scale by separating the sending and receiving of data. There are several important concepts to understand:

- Programs communicate by sending packets of data (messages) to each other.
- Channels (or queues) are logical pathways that connect the programs and convey messages.
- A sender or producer is a program that sends a message by writing the message to a channel.
- A receiver or consumer is a program that receives a message by reading it from a channel.

Important Messaging Concepts

- **Send and forget.** The sending application sends the message to the message channel (queue). Once that step is complete, the sender can move onto other work while the messaging system transmits the message in the background. The sender does not have to wait for the consumer to receive and process the message. There are ways to be notified once your intended recipient receives the message. "Read receipts"
- **Store and forward.** The messaging system stores the message, either in memory or on disk, and delivers the message to the receiver's computer.

Why Use a Messaging System?

Messaging capabilities are provided by a separate software system called a messaging system. A messaging system manages the channels that define the paths of communication between the applications and the sending and receiving of messages. The main task of a messaging system is to reliably move messages from the sender's computer to the receiver's computer.

The benefits of messaging include:

- **Remote communication.** Messaging enables separate applications to communicate and transfer data.

- **Platform/language integration.** When connecting multiple computer systems via remote communication, it's likely that these systems use different languages, technologies, and platforms. A messaging system allows these disparate to integrate. This is when multiple chat solutions try to integrate and pass the same information. This is also known as collaboration

- **Asynchronous communication.** Messaging enables a "send and forget" approach to communication. The sender does not have to wait for the receiver to receive and process the message.

- **Variable timing.** With synchronous communication, the caller must wait for the receiver to finish processing the call before the caller can receive the result and continue. This means that the caller can only make calls as fast as the receiver can perform them. Conversely, asynchronous communication allows the sender to batch requests to the receiver at its own pace and for the receiver to consume the requests at its own different pace.

- **Reliable communication.** The messaging uses a "store and forward" approach to transmitting messages. The data is packaged as messages, which are atomic, independent units. When the sender sends a message, the messaging system stores the message and then delivers it by forwarding the message to the receiver's computer. Not very much bandwidth is required for the sending/receiving of messages on any USMC network whether in the field or in garrison.

- **Topic-based messages.** Receivers can register to consume messages selectively based on particular topics of interest. Not ideal but available if required.



An Overview of XMPP Servers

XMPP is the Extensible Messaging and Presence Protocol, a set of open technologies for instant messaging, presence, multi-party chat, voice and video calls, collaboration, lightweight middleware, content syndication, and generalized routing of XML data.

XMPP was originally developed in the open-source community to provide an open, decentralized alternative to the closed instant messaging services at that time. XMPP offers several key advantages over such services:

- **Open** — the XMPP protocols are free, open, public, and easily understandable; in addition, multiple implementations exist in the form of clients, servers, server components, and code libraries.

- **Standard** — the Internet Engineering Task Force (IETF) has formalized the core XML streaming protocols as an approved instant messaging and presence technology. The XMPP specifications were published as RFC 3920 and RFC 3921 in 2004, and the XMPP Standards Foundation continues to publish many XMPP Extension Protocols.

- **Proven** — the first XMPP technologies were developed in 1998 and are now quite stable; hundreds of developers are working on these technologies, there are tens of thousands of XMPP servers running on the Internet today, and millions of people use XMPP for instant messaging through public services such as Google Talk and XMPP deployments at organizations worldwide.

- **Decentralized** — the architecture of the XMPP network is similar to email; as a result, anyone can run their own XMPP server, enabling individuals and organizations to take control of their communications experience. This is ideal for any unit that may be in a remote location and may be segregated from the main hub.

- **Secure** — any XMPP server may be isolated from the public network (e.g., on a company intranet) and robust security using SASL and TLS has been built into the core XMPP specifications.

- **Flexible** — XMPP applications beyond IM include network management, content syndication, collaboration tools, file sharing, gaming, remote systems monitoring, web services, lightweight middleware, cloud computing, and much more.

- **Diverse** — a wide range of companies and open-source projects use XMPP to build and deploy real-time applications and services

XMPP Servers that are used on USMC Networks

1. Cisco Jabber (Now called Cisco Webex)

Cisco Jabber combines presence information, instant messaging, voice and video communication, web conferencing and desktop sharing into a unified overall experience across mobile and stationary devices. Cisco Jabber is available for PC and Mac, as well as for tablets and smartphones. The key benefit of Jabber is that it is installed on premise therefore, if the link to the cloud goes down, it has no affect to users on the network. Jabber is currently being utilized by both I/IIMEF on their resident tactical networks.

Current features of Jabber available:

- **Instant Messaging (IM)** - Allows users to communicate instantly with people on their computers, mobile devices (notebooks and tablets) or smart phones.

- **Rich presence** - This allows users to see if a contact is available via Jabber, on the phone or in a meeting.

- **Desk phone integration** - Dial phone numbers via jabber client or dial phone numbers from within outlook or Jabber client.

- **Integrated directory** – Jabber has the ability to be linked with Active Directory giving users the ability to search for other people online.

- **Soft-phone** - Allows users to use their computer as a phone. Their computer would act as if it were an extra phone connected to their extension. This might be useful when there are not enough physical phones for every user on the network.

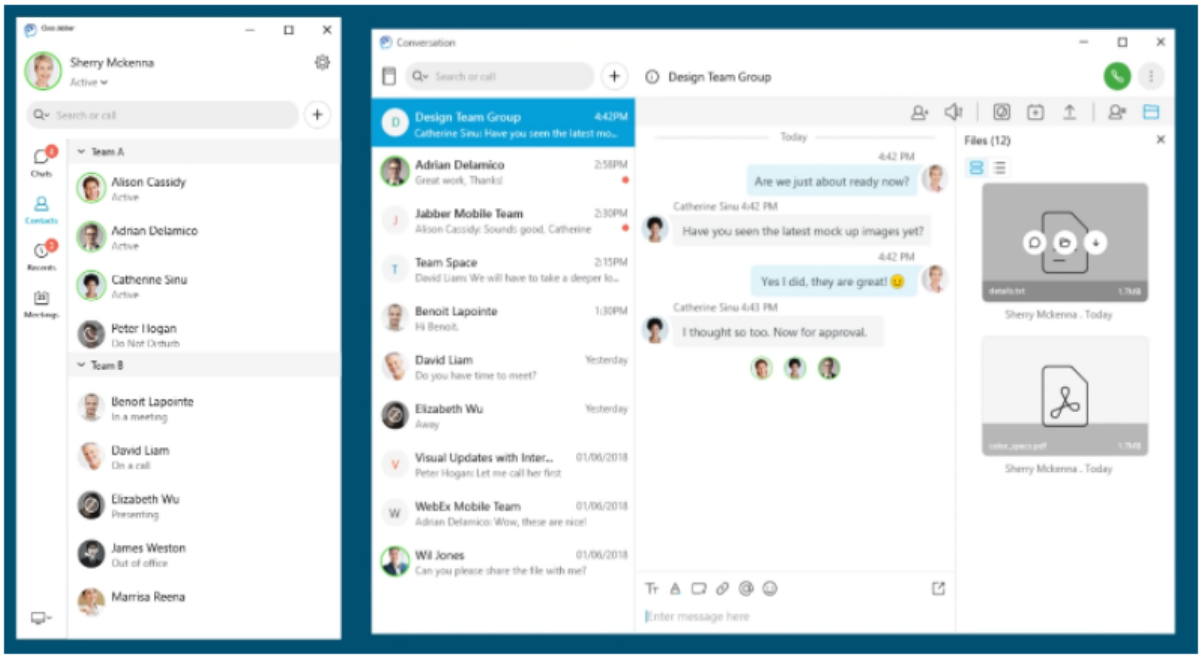
- **Visual voicemail** - Bring up user voicemail messages if enabled, they can see who called them, and listen to the messages.

- **Video Conferencing** - Video phone integrated with CDU video conferencing systems (compatible to all standards based video conferencing systems anywhere on the internet)

- **Desktop sharing** - Allow users to see what other users are seeing on their computer and allow them to interact with what they are seeing. This is extremely useful for Commanders when wanting to share information with other Commanders.

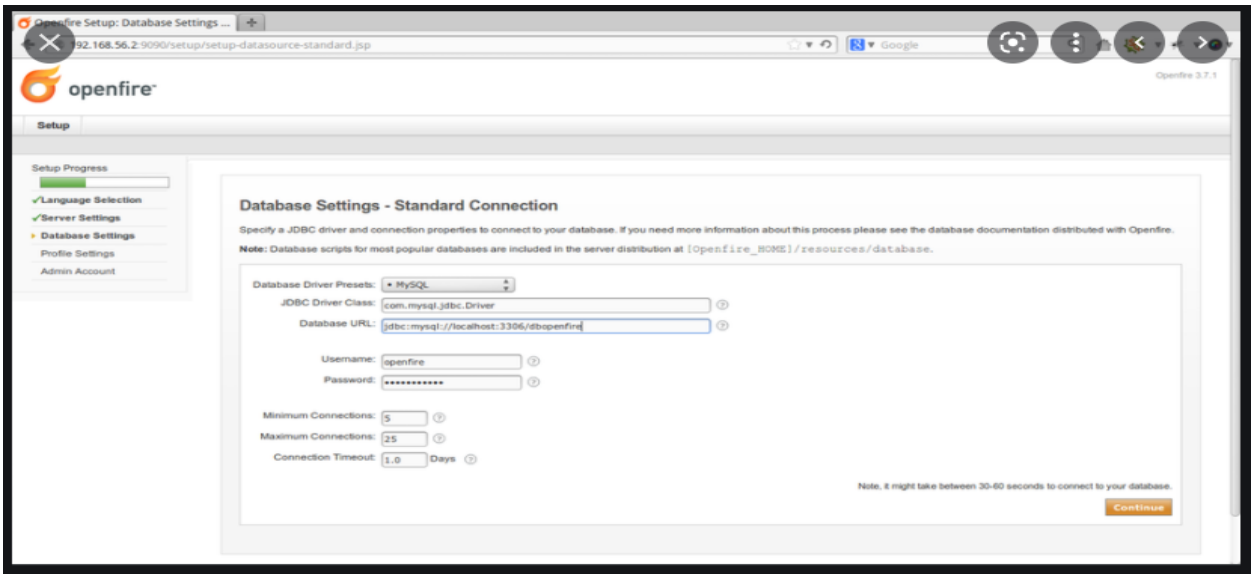
- **Collaborate with other organizations** - Most features extend to other institutions who also have Cisco Jabber installed. A good example is when I MEF was able to collaborate with both 1 MARDIV and MSTP.

- **Multiple Clients Available** - Clients for PC, Mac, iOS, Android. (Note: not not all features are available on all clients)



2. *Openfire*

Openfire is a real time collaboration (RTC) server licensed under the Open Source Apache License. It uses the only widely adopted open protocol for instant messaging, XMPP Openfire is incredibly easy to setup and administer, but offers rock-solid security and performance. Openfire it typically used in smaller networks where the amount of users is low however communication it still vital. Openfire typically resides on the Marine Corps Classified Networks.



Internet Relay Chat Servers (IRC)

Internet Relay Chat (IRC) is a text-based chat (instant messaging) system. It enables discussions among any number of participants in so-called conversation channels, as well as discussions between only two partners — for example, in question-and-answer dialogues. Any participant may open a new conversation channel, and a single computer user can also take part in several such simultaneous channels.

Internet Relay Chat is implemented as an application layer protocol to facilitate communication in the form of text. The chat process works on a client–server networking model. IRC clients can be stand-alone computer programs or web-based applications running either locally in the browser or on a third party server. These clients communicate with chat servers to transfer messages to other clients. IRC is mainly designed for group communication in discussion forums, called channels, but also allows one-on-one communication via private messages as well as chat and data transfer, including file sharing.

What are the benefits of using IRC?

- IRC channels allow real-time discussions among a large number of people at a time.
- Users can meet new people through the IRC channels.
- IRC chats can be logged if required which allows users to look at those logs in the future.
- There is no centralized control over IRC. Hence, any Marine can set up his/her own server and use it for their network.
- IRC has many access levels and complex channel modes. So, there is scope for privacy in form of restrictions in IRC. For example, few channels may need username/password to access, etc.

IRC Servers that are used on USMC Networks

1. Mako

MAKO supports requirements for low-bandwidth operations, making it ideal for in-theater submarines, airplanes and moving vehicles. MAKO servers are connected in a global federation, enabling server and clients to automatically reconnect with full chat history availability. There is web-based user administration across the federation simplifying administrative support of a distributed, federated server infrastructure. IRC does not seamlessly capture a record of war logging, while MAKO implements full requirements for recording war logging.

2. mIRC

mIRC is a full featured **Internet Relay Chat** client for Windows that can be used to communicate, share, play or work with others on IRC networks around the world, either in multi-user group conferences or in one-to-one private discussions.

It has a **clean, practical interface** that is highly configurable and supports features such as buddy lists, file transfers, multi-server connections, IPv6, SSL encryption, proxy support, UTF-8 display, UPnP, customizable sounds, spoken messages, tray notifications, message logging, and more.

mIRC also has a **powerful scripting language** that can be used both to automate mIRC and to create applications that perform a wide range of functions from network communications to playing games.

mIRC is a bit older however it was once used on USMC networks. Is it still used today? No but it could be an option.

Introduction to Programs of Record (POR)
What is a POR?

POR's are systems that come with pre-configured settings that are not allowed to be modified. They are to be deployed on any network and support any unit at any time. The unit responsible for providing support is called Marine Corps Tactical Systems Support Activity (MCTSSA).

MCTSSA conducts testing, evaluation, engineering, and provides direct technical support to the Fleet Marine Forces for Marine Corps and Joint Service command, control, computer, communications, intelligence (C4I) systems and expeditionary combat vehicle systems in order to inform acquisition decisions to make Marines more capable.

They provide the only Marine Corps organic, operationally relevant, MAGTF Command, Control, Communications, Computers, and Intelligence (C4I) architecture for enterprise-level testing, engineering, analysis, troubleshooting, and solutions. This environment directly supports investment decisions for the Objective Network by establishing a rigorous and repeatable process to evaluate proposed changes and cyber vulnerabilities from the tactical edge and throughout the Marine Corps Enterprise Network. MCTSSA enables Joint Communications Exercises, Joint Interoperability Testing, Tactical Cloud Implementation, Cyber Developmental Testing, Naval Integration, tactical networking/voice, and System of Systems testing. This environment is also the sole solution in the Marine Corps for Link-16 Tactical Data Link Networks for air-to-air, ship, and shore targeting and messaging, validated at MCTSSA before Marine/Navy use in theater.
The Warfighter Support Division is the section that helps most 0671's.

The Warfighter Support Division (WSD) provides global command, control, computers, communications and intelligence (C4I) systems support to the USMC, joint services, coalition partners, and other U.S. government agencies. We do this in order to ensure the warfighter can quickly, efficiently and intelligently operate their respective systems in the full scale of military operations.

Some examples of POR's include: Some of these POR's have chat solutions that 0671's need to be familiar with to support other organizations.

- ***Theater Battle Management Core System (TBMCS)***

The primary system for coordinating and executing the air campaign. It links the command and control systems for the United States Air Force, United States Navy, and United States Marine Corps, and integrates with ground systems for the U.S. Army systems, enabling coordinated, synchronized air battle management.

- ***Advanced Field Artillery Tactical Data System (AFATDS)***

AFATDS supports weapon systems such as mortars, field artillery cannons, rockets, close air support, attack helicopters, and Naval Surface Fire Support (NSFS) systems.

- ***Command and Control Personal Computer (C2PC)***

Developed for the U.S. Marines Corps, C2PC displays the Common Operational Picture (COP) across multiple workstation and among multiple agencies.

- ***Joint Tactical Common Operational Picture Workstation (JTCW)***

It is a single digital display of relevant operational information shared by battalion and higher leadership connected on the Global Combat and Control System Tactical Combat Operations network. It combines seven tactical applications into one user interface.

- ***Combat Data Network (CDN) Small Form Factor (SFF)***

The United States Marines Corps (USMC) Combat Data Network (CDN) Small Form Factor (SFF) program integrates Commercial Off-The-Shelf (COTS) and Government Off-The-Shelf (GOTS) software and hardware into deployable, modular, environmentally resilient, scalable suite of computer network equipment.

**MCTSSA Helpful Resources*

MCTSSA Warfighter Support Division
https://www.marcorsyscom.marines.mil/MCTSSA/Divisions/Warfighter-Support-Division

MCTSSA Support Center
https://hcs.usmc.mil/sites/mctssa/Pages/Home.aspx#/solutions

MARCORSYSCOM Portfolios and Programs
https://www.marcorsyscom.marines.mil/Portfolios-and-Programs/

Microsoft Exchange

What is Microsoft Exchange?

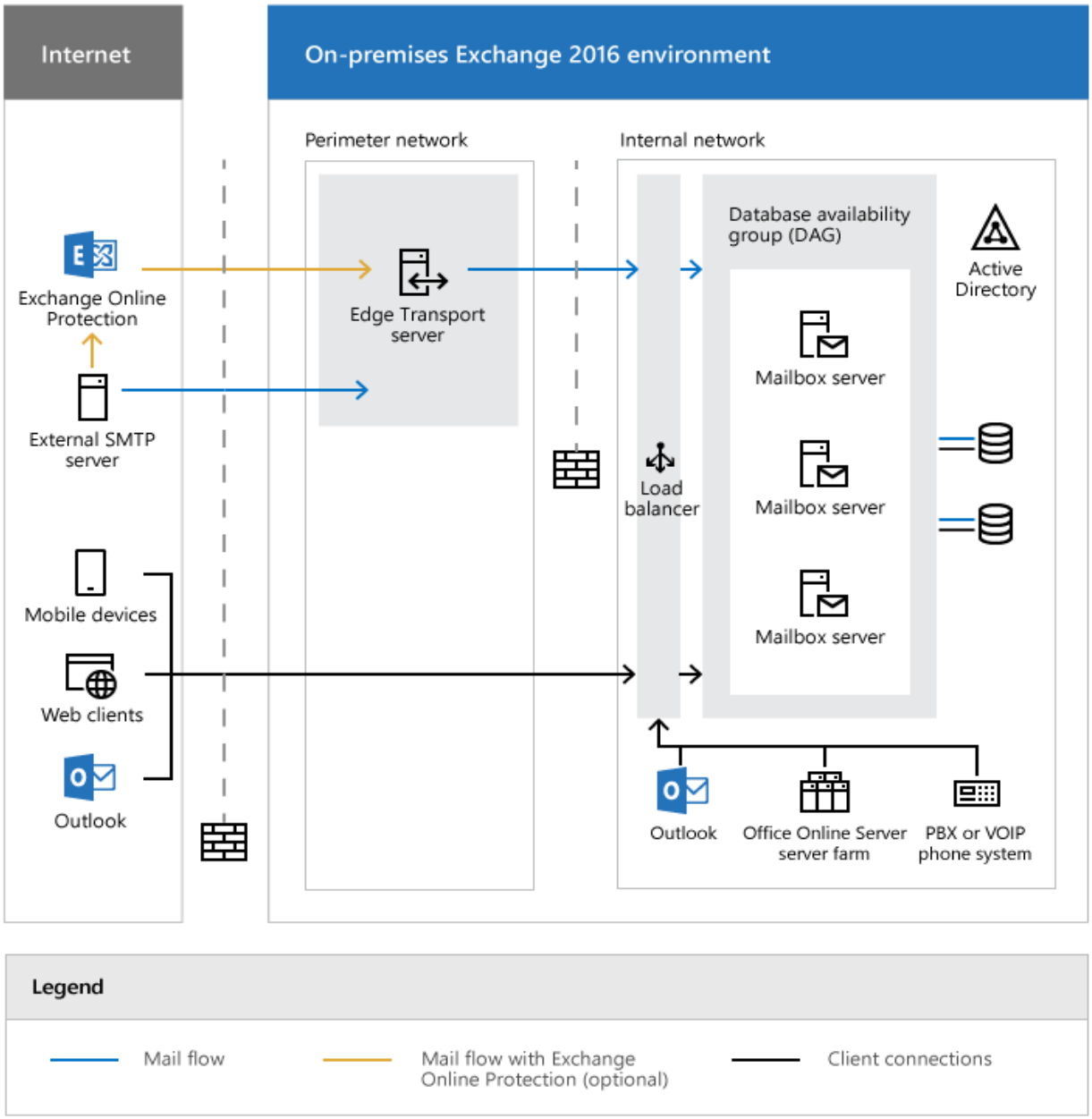
Microsoft Exchange is “a collection of applications that enable digital messaging and collaboration in an enterprise IT environment.” Exchange is a mail and calendar server, which means it is a dedicated network resource management program that uses transmission control protocols such as SMTP, IMAP and POP to communicate with email clients. In simpler terms, Exchange is responsible for sending and receiving emails to and from client computers. Exchange can be paired with any email client, though it is most commonly used in conjunction with Microsoft Outlook. Exchange is included Microsoft Office 365 for Business, the company’s collaboration and productivity suite. Exchange is used on almost every USMC network whether in garrison or on tactical.

How Microsoft Exchange works

Microsoft Exchange relies on the use of an Exchange server — a computer on which individual users' Exchange accounts are configured. An organization can maintain its own Exchange server or rely on Microsoft to do that via the cloud using a Microsoft 365 account.
When administrators use a Microsoft Exchange account, email messages — along with calendar information and other Outlook details — are generally kept in sync between their Outlook client on a computer or mobile device and the Exchange server. This most often happens using a technology called Exchange ActiveSync. ActiveSync keeps all the Exchange data synchronized between devices, so when email is replied to, deleted, or moved on their computer, for example, that change is immediately made on the Exchange server as well, and then synchronized with any other devices users might use, such as Outlook on a mobile device.

ActiveSync isn't the only option, though; an organization can choose to manage its email using IMAP or POP as well. IMAP is very similar to ActiveSync in that it keeps email messages in sync between the Exchange server and clients, though POP works differently — it downloads email from the Exchange server to a single computer and does not keep the two devices in sync. It's not commonly used for Exchange systems.

Exchange use a single building block architecture that provides email services for deployments at all sizes, from small organizations to the largest multi-national corporations. This architecture is describe in the following diagram.



Server Communication Architecture

Communication between Exchange servers and past and future versions of Exchange occurs at the protocol layer. Cross-layer communication isn't allowed. This communication architecture is summarized as "every server is an island". This architecture has the following benefits:

- Reduced inter-server communications.
- Version-aware communications.
- Isolated failures.
- Integrated design inside each server.

Server role architecture

Mailbox servers

- Mailbox servers contain the transport services that are used to route mail.
- Mailbox servers contain mailbox databases that process, render, and store data.
- Mailbox servers contain the Client Access services that accept client connections for all protocols. These frontend services are responsible for routing or proxying connections to the corresponding backend services on a Mailbox server. Clients don't connect directly to the backend services.
- Administrators manage Mailbox servers by using the Exchange admin center (EAC) and the Exchange Management Shell.

Edge Transport servers

- Edge Transport servers handle all external mail flow for the Exchange organization.
- Edge Transport servers are typically installed in the perimeter network, and are subscribed to the internal Exchange organization.
- Edge Transport servers provide antispam and mail flow rules as mail enters and leaves your Exchange organization.
- Administrators can manage Edge Transport servers by using the Exchange Management Shell.

Characteristics of Messaging Servers

A good messaging server architecture quickly delivers email with embedded sound, graphics, video files, and HTML forms, while providing for future upgrade and scalability. At a simplistic level, the Messaging Server architecture should:

- Accept incoming mail from external sites
- Determine the user mailbox to deliver these messages to and route them accordingly
- Accept incoming mail from internal hosts
- Determine the destination system to deliver these messages to and route them accordingly

Central to a messaging system architecture is the messaging server itself, a collection of components used to send and deliver messages. In addition to components provided in Messaging Server, the email system also requires an LDAP server and a DNS server. The DNS server must be in place before deploying the messaging system.

The Requirements of Messaging Servers on a Domain

MX Records

The MX records for the domain are the most important, as they are what the rest of the Internet uses to know where to send email for the perspective domain.
The MX record is set with the domain name management provider.
An MX record is made up of two parts.

- The host. This will be something along the lines of mail.example.com and will point to an IP address.
- The MX record itself. This points to a host. Pointing the MX record to an IP address is not valid. While it will work in some cases, administrators will get failures.

  However the name used on the host does play a part with the other DNS configurations made.

The MX weight or priority indicates which server should be used first. The lowest number is tried first. This is usually incremented in units of five. Five is the lowest, followed by ten, fifteen etc. It is very rare to see the number one.

Reverse DNS

Reverse DNS is also important. In many cases the reverse DNS information is used to check that the exchange server is who it says it is. The RDNS record (also known as PTR) is attached to the IP address. As such, only one Reverse DNS per IP address.

The reverse DNS should ideally match the MX record, and what the server has on the SMTP banner when connecting.

Service Location (SRV)

The SRV record is a Domain Name System (DNS) resource record. It's used to identify computers hosting specific services. SRV resource records are used to locate domain controllers for Active Directory. *These are primarily used for chat servers that are on the domain.

Autodiscover DNS Record

Autodiscover is a service that automatically configures email clients, such as Outlook. It is a necessary requirement for correct access and smooth operation of the Hosted Exchange service.

Data Availability Groups (DAG's)

A database availability group (DAG) is a set of up to 16 Exchange Mailbox servers that provides automatic, database-level recovery from a database, server, or network failure. DAGs use continuous replication and a subset of Windows failover clustering technologies to provide high availability and site resilience. Mailbox servers in a DAG monitor each other for failures. When a Mailbox server is added to a DAG, that server works with the other servers in the DAG to provide automatic, database-level recovery from database failures.

When a DAG is created, it's initially empty. When administrators add the first server to a DAG, a failover cluster is automatically created for the DAG. In addition, the infrastructure that monitors the servers for network or server failures is initiated. The failover cluster heartbeat mechanism and cluster database are then used to track and manage information about the DAG that can change quickly, such as database mount status, replication status, and last mounted location.

Exchange administrators can choose to use DAGs in different geographic locations and in many different architectural designs and configurations, making it considerably more flexible than previous high-availability setups.

How does it work?

The DAG replicates the mailbox databases between the mailbox servers. The more servers that are included, the more copies can be shared throughout the DAG group. As typical for a cluster, it also contains a heartbeat, cluster networks, and the cluster database.
The DAG group always ***has one*** active server. The rest are set on passive. That means, depending on the structure that is setup:

- The mailbox databases are spread across multiple DAG members --> that ensures that no two servers have the same mix of databases.
- The databases of the active server are replicated to the passive server --> direct copy of the active server
- The DAG replicates the data on a remote server --> also called site resilience, as it guarantees a 'remote copy' of the data

DAG quorum modes

A DAG is a specific implementation of a Windows Server Cluster. The Exchange components of DAGs rely on the underlying Windows Server Cluster technology to work. The concept of quorum is essential to understanding how to implement and manage DAGs.

Quorum is the idea that, in the event of a failure of some DAG members, there are rules to govern what resources the remaining members can provide. These quorum rule sets exist to provide consistent operation of a DAG and act as a tiebreaker in situations where DAG nodes lose communication with each other.

When a DAG has an even number of nodes, it uses Node & File Share Majority quorum mode. In this mode, an external witness server acts as the tiebreaker. When running in this mode, each DAG node member gets a single vote, but the witness server gives one of the DAG nodes an additional vote. The cluster quorum data is stored on each member's local system disk, but the witness server has a separate file that points to one DAG member as the most updated copy of the DAG cluster quorum data.

When a DAG has an odd number of members, it uses Node Majority quorum mode. In this mode, each DAG member gets a vote, and each member's local system disk is used to store cluster quorum data.

It is possible to manually assign specific DAG members with weighted quorum votes. Doing so is not recommended in most circumstances and should only be done after direct consultation with Microsoft support.

The requirements for the witness server are as follows:

- The witness server can't be a member of the DAG.
- The witness server must be in the same Active Directory forest as the DAG.
- The witness server must be running Windows Server 2008 or later.
- A single server can serve as a witness for multiple DAGs. However, each DAG requires its own witness directory.

High Availability Solutions

What is High Availability?

A solution that provides service availability, data availability, and automatic recovery from failures that affect the service or data (such as a network, storage, or server failure).

**DAG's, previously covered are the solution for exchange.*

What about Chat servers or other types of messaging?

Cisco Jabber: Cisco Jabber supports a three-server cluster. Three operational servers are required for full redundancy. Cisco Jabber will function properly with two operational servers in the cluster; however, it is recommend to only run with two operational servers for short periods of time, for example, when upgrading the servers in a Cisco Jabber cluster.

Mako: The Mako VPN Cloud SD-WAN solution from Mako Networks can be configured to provide Fast Failover, a High Availability (HA) feature that enables real-time, automated failover from one broadband network to another to take place without disrupting service delivery. This means critical ogranizations systems stay online and do not drop a packet during unexpected network outages. Already known for world-class security features, the seamless performance of Fast Failover sets Mako apart from the competition. The likelihood of this concept being deployed is slim to none however if needed the solution is identified.

Openfire Server: As of today, there is no use case where Openfire servers have deployed in HA mode. If it were to be deployed in such a scenario, this would be over a distributed network where users were separated.

Exchange Server PowerShell (Exchange Management Shell)

What is Exchange Management Shell?

The Exchange Management Shell is built on Windows PowerShell technology and provides a powerful command-line interface that enables the automation of Exchange administration tasks. Administrators can use the Exchange Management Shell to manage every aspect of Exchange. For example, they can create email accounts, create Send connectors and Receive connectors, configure mailbox database properties, and manage distribution groups. Administrators can use the Exchange Management Shell to perform every task that's available in the Exchange graphical management tools, plus things that they can't do there (for example, bulk operations). In fact, when administrators do something in the Exchange admin center (EAC), the Exchange Control Panel (ECP), or the Exchange Management Console (EMC), it's the Exchange Management Shell that does the work behind the scenes.

The Exchange Management Shell also provides a robust and flexible scripting platform. Visual Basic scripts that required many lines of code can be replaced by Exchange Management Shell commands that use as little as one line of code. The Exchange Management Shell provides this flexibility because it uses an object model that's based on the Microsoft .NET Framework. This object model enables Exchange cmdlets to apply the output from one command to subsequent commands.

During the lab portion, students will have the opportunity to utilize Exchange Management Shell to see some of the capabilities.

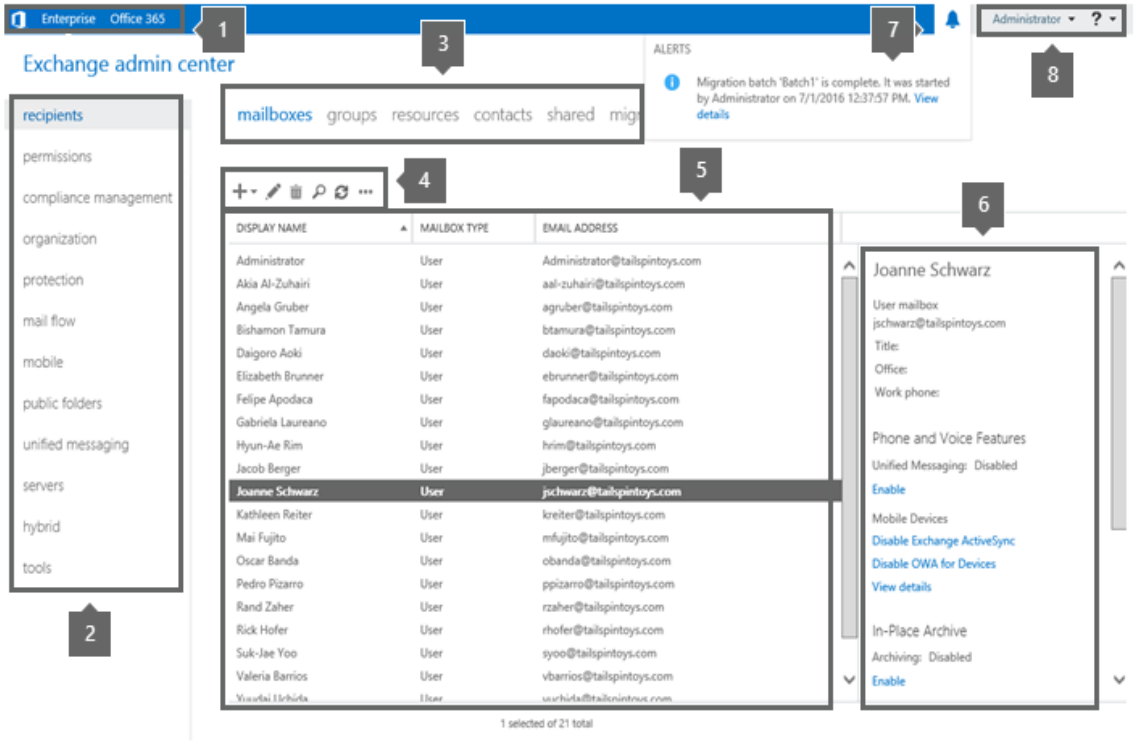
Exchange admin center in Exchange Server

The Exchange admin center (EAC) is the web-based management console in Exchange Server that's optimized for on-premises, online, and hybrid Exchange deployments. The EAC was introduced in Exchange Server 2013, and replaces the Exchange Management Console (EMC) and the Exchange Control Panel (ECP), which were the two management interfaces in Exchange Server 2010.

This is the GUI portion of administering Exchange, rather than use Exchange Management Shell.

*The URL of the EAC is controlled by the Internet Information Services (IIS) virtual directory named ECP in the Client Access (frontend) services on the Mailbox server. Yes, the virtual directory is named ECP, not EAC.

The section describes the user interface elements that are common across the EAC.



Messaging Server Client Applications

Microsoft Outlook Client

Microsoft Outlook is the email client included with the Microsoft Office suite. It is designed to operate as an independent personal information manager, as an Internet mail client, or in conjunction with the Microsoft Exchange Server for group scheduling, email, and task management. It manages email, calendars, contacts, tasks, to-do lists, and documents or files on the hard drive. Outlook helps users communicate through email, phone support, and group scheduling capabilities. Outlook also helps individuals share information by means of public folders, forms, and Internet connectivity. Users can access their mailbox by both the Outlook Client and Outlook Web Access portal.



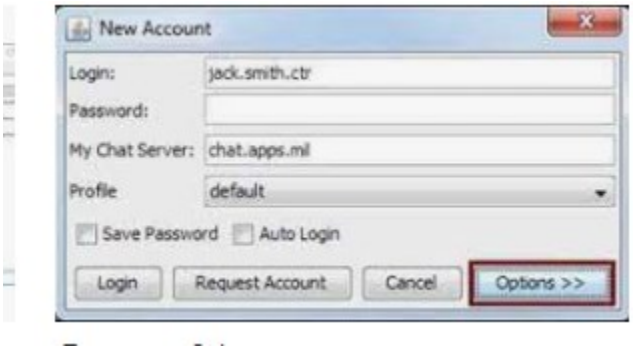
Cisco Jabber Client

Cisco Jabber is a communication tool that allows users to view others' availability status and access instant messaging (IM), voice, video, voice messaging, desktop sharing, and conferencing.



Transverse Client

Transverse chat client is a communication tool that allows users to view others' availability status and access instant messaging (IM), desktop sharing, and conferencing. It is used to connect to an Openfire server. Transverse can be both Active Directory integrated or used for local login capabilities.



Spark Client

Spark is an Open Source, cross-platform IM client optimized for businesses and organizations. It features built-in support for group chat, telephony integration, and strong security. It also offers a great end-user experience with features like in-line spell checking, group chat room bookmarks, and tabbed conversations.

